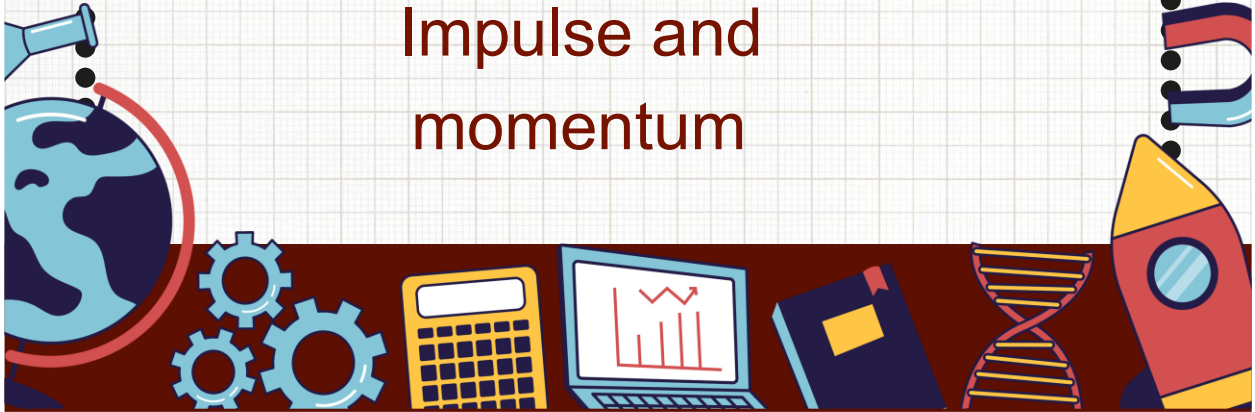


GRADE 11 - GENERAL SCIENCE

LESSON 3

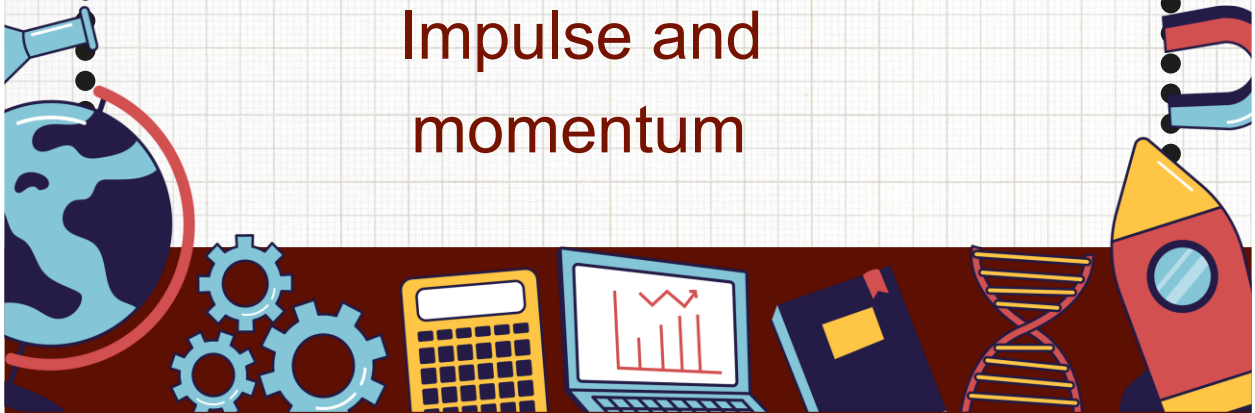
Impulse and momentum



GRADE 11 - GENERAL SCIENCE

LESSON 3

Impulse and momentum



MOMENTUM

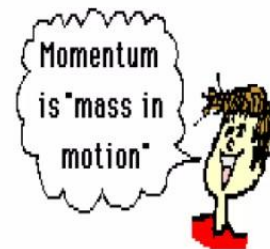
Linear Momentum or Simply Momentum is the product of the mass and velocity of an object. Like velocity, linear momentum is a vector quantity, possessing a direction as well as a magnitude:

$$P = mv.$$

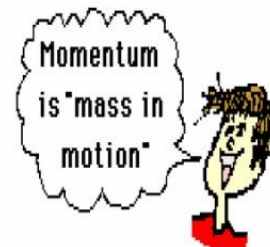
$$\text{Momentum} = \text{mass} \cdot \text{velocity}$$

Using P as a symbol for momentum. The SI unit for momentum is kg m/s.

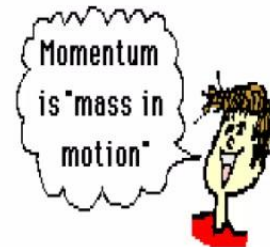
- All objects have mass; so if an object is moving, then it has momentum - it has its mass in motion.
- The amount of momentum which an object has is dependent upon two variables:
 - how much matter is moving?
 - how fast the matter is moving?



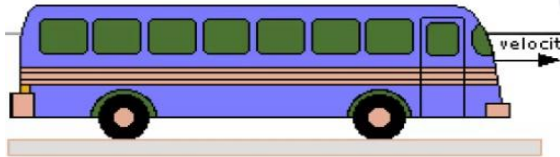
- All objects have mass; so if an object is moving, then it has momentum - it has its mass in motion.
- The amount of momentum which an object has is dependent upon two variables:
 - how much matter is moving?
 - how fast the matter is moving?



- All objects have mass; so if an object is moving, then it has momentum - it has its mass in motion.
- The amount of momentum which an object has is dependent upon two variables:
 - how much matter is moving?
 - how fast the matter is moving?



A bus can have a **large momentum** even if it is moving very slowly, because it has a **large mass**.



$$(\text{mass})(\text{velocity}) = \text{momentum}$$

A bullet can have a **large momentum** even if it has a small mass, because it is moving at **high velocity**.



$$(\text{mass})(\text{velocity}) = \text{momentum}$$

Momentum and Inertia

- Inertia is another property of mass that resists changes in velocity; however, inertia depends only on mass.
- Inertia is a scalar quantity.
- Momentum is a property of moving mass that resists changes in a moving object's velocity.
- Momentum is a vector quantity.

Momentum and Inertia

- Inertia is another property of mass that resists changes in velocity; however, inertia depends only on mass.
- Inertia is a scalar quantity.
- Momentum is a property of moving mass that resists changes in a moving object's velocity.
- Momentum is a vector quantity.



Momentum Questions

1. Determine the momentum of a ...
 - a.) 60 kg halfback moving eastward at 9 m/s.
 $p = 540 \text{ kg}\cdot\text{m/s}$, east
 - b.) 1000 kg car moving northward at 20 m/s.
 $p = 20,000 \text{ kg}\cdot\text{m/s}$, north
 - c.) 40 kg man moving southward at 2 m/s.
 $p = 80 \text{ kg}\cdot\text{m/s}$, south



Change in Momentum

- Newton's second law states that *the net external force acting on an object is equal to the time rate of change of the object's momentum.*

$$F_{net} = \frac{\Delta p}{\Delta t}$$



Change in Momentum

- Newton's second law states that *the net external force acting on an object is equal to the time rate of change of the object's momentum.*

$$F_{net} = \frac{\Delta p}{\Delta t}$$



Change in Momentum

- Newton's second law states that *the net external force acting on an object is equal to the time rate of change of the object's momentum.*

$$F_{net} = \frac{\Delta p}{\Delta t}$$



Force is the Rate of Change of Momentum

- Momentum changes when a net force is applied.
- The inverse is also true:
 - If momentum changes, forces are created.
- If momentum changes quickly, large forces are involved.

- These concepts are merely an outgrowth of Newton's Second law as discussed in the previous lesson.
- Newton's second law ($F_{\text{net}} = m \cdot a$) stated that the acceleration of an object is directly proportional to the mass of the object. When combined with the definition of acceleration ($a = \text{change in velocity/time}$), the following equalities result.

$$F = m \cdot a = m \cdot \frac{\Delta v}{t}$$

or

$$F = m \cdot \frac{\Delta v}{t}$$

- If both sides of the above equation are multiplied by the quantity t , a new equation results.

$$F \cdot t = m \cdot \Delta v$$

In physics, the quantity Force $\cdot$ time is known as **impulse**. And since the quantity $m \cdot v$ is the momentum, the quantity $m \cdot \Delta v$ must be the **change in momentum**. The equation really says that the
Impulse = Change in momentum

Impulse-Momentum Theorem

The impulse due to all forces acting on an object (the net force) is equal to the change in momentum of the object:

$$\mathbf{F}_{\text{net}} t = \Delta \mathbf{p}$$

We know the units on both sides of the equation are the same

(last slide), but let's prove the theorem formally:

$$\mathbf{F}_{\text{net}} t = m a t = m (\Delta \mathbf{v} / t) t = m \Delta \mathbf{v} = \Delta \mathbf{p}$$

Impulse-Momentum Theorem

The impulse due to all forces acting on an object (the net force) is equal to the change in momentum of the object:

$$\mathbf{F}_{net} t = \Delta \mathbf{p}$$

We know the units on both sides of the equation are the same

(last slide), but let's prove the theorem formally:

$$\mathbf{F}_{net} t = m a t = m (\Delta \mathbf{v} / t) t = m \Delta \mathbf{v} = \Delta \mathbf{p}$$

Impulse-Momentum Theorem

The impulse due to all forces acting on an object (the net force) is equal to the change in momentum of the object:

$$\mathbf{F}_{net} t = \Delta \mathbf{p}$$

We know the units on both sides of the equation are the same

(last slide), but let's prove the theorem formally:

$$\mathbf{F}_{net} t = m a t = m (\Delta \mathbf{v} / t) t = m \Delta \mathbf{v} = \Delta \mathbf{p}$$

Impulse-Momentum Theorem

The impulse due to all forces acting on an object (the net force) is equal to the change in momentum of the object:

$$\mathbf{F}_{net} t = \Delta \mathbf{p}$$

We know the units on both sides of the equation are the same

(last slide), but let's prove the theorem formally:

$$\mathbf{F}_{net} t = m a t = m (\Delta \mathbf{v} / t) t = m \Delta \mathbf{v} = \Delta \mathbf{p}$$

Impulse

- The product of a force and the time the force acts is called the *impulse*.
- Impulse is a way to measure a *change in momentum* because it is not always possible to calculate force and time individually since collisions happen so fast.

Impulse

- The product of a force and the time the force acts is called the *impulse*.
- Impulse is a way to measure a *change in momentum* because it is not always possible to calculate force and time individually since collisions happen so fast.

- A change in momentum in a short time requires a large force.
- A change in momentum in a long time requires a small force.



Force and Momentum Change

To find the impulse, you rearrange the momentum form of the second law.

$$\text{Impulse (N}\cdot\text{sec)} \longrightarrow \vec{F} \Delta t = \Delta \vec{p}$$

Impulse can be expressed in kg•m/sec (momentum units) or in N•sec.

Change in momentum (kg•m/sec)

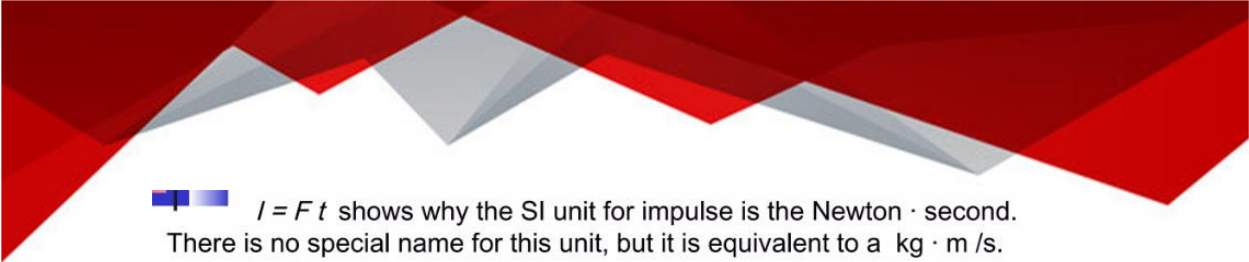
Impulse Defined

Impulse is defined as the product force acting on an object and the time during which the force acts. The symbol for impulse is I . So, by definition:

$$I = F t$$

Example: A 50 N force is applied to a 100 kg boulder for 3 s. The impulse of this force is $I = (50 \text{ N}) (3 \text{ s}) = 150 \text{ N}\cdot\text{s}$.

Note that we didn't need to know the mass of the object in the above example.



 $I = F t$ shows why the SI unit for impulse is the Newton · second. There is no special name for this unit, but it is equivalent to a $\text{kg} \cdot \text{m} / \text{s}$.

proof: $1 \text{ N} \cdot \text{s} = 1 (\text{kg} \cdot \text{m} / \text{s}^2) (\text{s}) = 1 \text{ kg} \cdot \text{m} / \text{s}$

$F_{\text{net}} = m a$ shows this
is equivalent to a newton.

Therefore, impulse is equal to momentum but different in units, which leads to a useful theorem.